

UNIT THREE

MAKING WELL-INFORMED FOOD CHOICES

- **UNIT OBJECTIVES**

- By the end of this unit students should be able to:-
 - Summarize the basics of nutrition
 - Define macronutrients and micronutrients
 - Categorize and describe the body's sources of energy
 - Analyse intake of nutrition and make appropriate changes for sound eating practices
 - Realize the management of nutrition for better physical performance

Introduction

- Regular exercise taxes every cell and every system in your body: your muscles, joints, ligaments, tendons, respiratory, circulatory and immune systems all have to work harder when you exercise.
- Eating a healthy diet can help minimize the damage caused by exercise and help your body rebuild itself even stronger.

Sound Eating Practices

- What is Nutrition? And
- what is Nutrient?
- Nutrition can be defined as the science of the action of food, beverages, and their components in biological systems.
- A nutrient is a compound that provides a needed function in the body.
- Nutrients can be further classified based on the amount needed in the body.
- Macronutrients are the nutrients the body needs in larger amounts.
- Micronutrients are also important nutrients, but ones the body needs in smaller amounts.

- The food you eat is a source of nutrients. Nutrients are defined as the substances found in food that keep your body functioning.
- Your body needs nutrients to...
- Fuel your energy.
- Help you grow.
- Repair itself.
- Maintain basic bodily functions

- **Carbohydrates:** The word carbohydrate literally means "hydrated carbon," or carbon with water. Thus, it is no surprise that carbohydrates are made up of carbon, hydrogen, and oxygen. Sucrose (table sugar) is an example of a commonly consumed carbohydrate. Some dietary examples of carbohydrates are whole-wheat bread, oatmeal, rice, sugary snacks/drinks, and pasta.

- **Proteins:** Like carbohydrates, proteins are comprised of carbon, hydrogen, and oxygen, but they also contain nitrogen. Several dietary sources of proteins include nuts, beans/legumes, skim milk, egg whites, and meat.

Function in the Body:

Provides energy.

Help to build, maintain, and repair body tissues.

Proteins are made up of chemical compounds called amino acids. There are 20 amino acids.

Lipids: Lipids consist of fatty acids, triglycerides, phospholipids, and sterols (cholesterol). Lipids are also composed of carbon, hydrogen, and oxygen. Some dietary sources of lipids include oils, butter, and egg yolks.

- **Water:** Water is made up of hydrogen and oxygen and is the only macronutrient that provides no energy.

Micronutrients

- Vitamins: These compounds are essential for normal physiologic processes in the body.

Unlike carbohydrates, fats, and proteins, vitamins DO NOT provide energy (calories).

- Minerals: Minerals are the elements (think periodic table) that are essential for normal physiologic processes in the body.

Unlike carbohydrates, fats, and proteins, minerals DO NOT provide energy (calories).

- Food energy is measured in kilocalories (kcal), commonly referred to as calories. Although technically incorrect, this terminology is so familiar that it will be used throughout this course.
- A kilocalorie is the amount of energy needed to raise 1 kilogram of water 1 degree Celsius. A food's kilocalories are determined by putting the food into a bomb calorimeter and determining the energy output:
- Energy = Measurement of Heat Produced. Below is a picture of a bomb calorimeter and a link to a video showing how one is used. The number of kilocalories per gram for each nutrient is shown below:

Amount of calories obtained from nutrients

<i>Nutrient</i>	<i>Energy (kcal/g)</i>
<i>Carbohydrate</i>	<i>4</i>
<i>Protein</i>	<i>4</i>
<i>Lipids</i>	<i>9</i>
<i>Water</i>	<i>0</i>
<i>Vitamins</i>	<i>0</i>
<i>Minerals</i>	<i>0</i>

carbohydrates, protein, and lipids provide energy. However, there is another dietary energy source that is not a nutrient— alcohol. To emphasize, alcohol is not a nutrient, but it does provide 7 kilocalories of energy per gram.

daily calorie needs will depend on your genetic make-up, age, weight, body composition, your daily activity and your training programme. It is possible to estimate the number of calories you need daily from your body weight (BW) and your level of daily physical activity.

Step 1: Estimate your Basal Metabolic Rate (BMR)

As a rule of thumb, BMR uses 22 calories for every kg of a woman's body weight and 24 calories per kg of a man's body weight.

Women: $\text{BMR} = \text{weight in kg} \times 22$

Men: $\text{BMR} = \text{weight in kg} \times 24$

Step 2: Work out your Physical Activity Level (PAL)

This is the ratio of your overall daily energy expenditure to your BMR; a rough measure of your lifestyle activity.

- Mostly inactive or sedentary (mainly sitting): 1.2
- Fairly active (include walking and exercise 1–2 x week): 1.3
- Moderately active (exercise 2–3 x weekly): 1.4
- Active (exercise hard more than 3 x weekly): 1.5
- Very active (exercise hard daily): 1.7

Step 3: Multiply your BMR by your PAL to work out your Daily Calorie Needs

- Daily calorie needs = $\text{BMR} \times \text{PAL}$
- This figure gives you a rough idea of your daily calorie requirement to maintain your weight. If you eat fewer calories, you will lose weight; if you eat more then you will gain weight.

I. Carbohydrates

- Carbohydrate is an important fuel for exercise. It is stored as glycogen in your liver and muscles, and must be re-stocked each day.
- Approximately 100 g glycogen (equivalent to 400 kilocalories) may be stored in the liver, and up to 400 g glycogen (equivalent to 1600 kilocalories) in muscle cells.
- The purpose of liver glycogen is to maintain steady blood sugar levels.

- The more active you are, the higher your carbohydrate needs. Guidelines for daily intakes are about 5–7 g per kg of body weight per day for moderate duration/low intensity daily training.
- Those who do moderate–heavy endurance training should consume 7–10 g per kg body weight per day; and those training more than 4 hours per day are advised to consume 10 g or more per kg body weight per day.

- During exercise lasting longer than 60 minutes, consuming 20–60 g carbohydrate per hour helps maintain your blood glucose level, delay fatigue and increase your endurance. Choose high GI carbohydrates (e.g. sports drinks, energy gels and energy bars, bananas, fruit bars, cereal or breakfast bars), which convert into blood sugar rapidly.

II. Protein

Protein is another major macronutrient that, like carbohydrates, consists of small repeating units. But instead of sugars, proteins are made up of amino acids.

Proteins can be classified as either complete or incomplete. Amino acids from proteins form the building blocks for new tissues and the repair of body cells.

They are also used for making enzymes, hormones and antibodies.

Protein also provides a (small) fuel source for exercising muscles.

- As protein is found in so many foods, most people, including athletes eat a little more protein than they need.
- This isn't harmful, the excess is broken down into urea (which is excreted) and fuel, which is either used for energy or stored as fat if your calorie intake exceeds your output.
- Several studies have found that carbohydrate and protein eaten together immediately after exercise enhances recovery and promotes muscle building

III. Fat

- Some fat is essential, it makes up part of the structure of all cell membranes, your brain tissue, nerve sheaths, bone marrow and it cushions your organs.
- Fat in food also provides essential fatty acids, the fat-soluble vitamins A, D and E, and is an important source of energy for exercise.
- about 20–33% of the calories in your diet should come from fat.

- ‘Bad’ fats (saturated and trans fats) should be kept to a minimum (the UK government recommends less than 10% of calories), with the majority coming from ‘good’ (unsaturated) fats.
- Omega-3s may be particularly beneficial for athletes as they help increase the delivery of oxygen to muscles, improve endurance and may speed recovery, reduce inflammation and joint stiffness.
- There is neither bad nor good cholesterol, despite the common use of these descriptions in reference to LDL and HDL, respectively.
- Cholesterol is cholesterol. HDL and LDL contain cholesterol but are actually lipoproteins

- The simplest definition of fiber is indigestible matter. Indigestible means that it survives digestion in the small intestine and reaches the large intestine. There are the three major fiber classifications:
 - Dietary fiber: This type of fiber contains both non-digestible carbohydrates and lignin and is always intrinsic and intact in plants.
 - Functional fiber: This type of fiber contains non-digestible carbohydrates only and can be isolated, extracted, or synthesized. Functional fiber can be from plants or animals and produces beneficial physiological effects in humans.
 - Total Fiber: Fiber that contains both dietary fiber and functional fiber.

- Vitamins are organic compounds found in foods and are a necessary part of the biochemical reactions in the body.
- They are involved in a number of processes, including mineral and bone metabolism, and cell and tissue growth, and they act as cofactors for energy metabolism. The B vitamins play the largest role of any vitamins in metabolism.
- Vitamins are either fat-soluble or water-soluble. Fat-soluble vitamins A, D, E, and K, are absorbed through the intestinal tract with lipids in chylomicrons.
- Vitamin D is also synthesized in the skin through exposure to sunlight.

- Water-soluble vitamins, including the eight B vitamins and vitamin C, are absorbed with water in the gastrointestinal tract.
- These vitamins move easily through bodily fluids, which are water based, so they are not stored in the body. Excess water-soluble vitamins are excreted in the urine.

- Minerals in food are inorganic compounds that work with other nutrients to ensure the body functions properly.
- Minerals cannot be made in the body; they come from the diet
- most of that consists of the minerals that the body requires in moderate quantities: potassium, sodium, calcium, phosphorus, magnesium, and chloride.

- The most common minerals in the body are calcium and phosphorous, both of which are stored in the skeleton and necessary for the hardening of bones.
- Most minerals are ionized, and their ionic forms are used in physiological processes throughout the body.

- You should ensure you are hydrated before starting training competition and aim to minimize dehydration during exercise.
- Dehydration can result in reduced endurance and strength, and heat related illness.
- The IOC advises matching your fluid intake to your fluid losses as closely as possible and limiting dehydration to no more than 2% loss of body weight (e.g. a body weight loss of no more than 1.5 kg for a 75 kg person).

- Additionally, the IAAF cautions against overhydrating yourself before and during exercise, particularly in events lasting longer than 4 hours. Constantly drinking water may dilute your blood so that your sodium levels fall.
- Sports drinks are better than water during intense exercise lasting more than 60 minutes because their sodium content will promote water retention and prevent hyponatraemia.

Nutrition and Physical Performance

- There is universal scientific consensus that diet affects performance. A well-planned eating strategy will help support any training programme.
- everyone has different nutritional needs and there is no single diet that fits all.
- Some athletes require more calories, protein or vitamins than others; and each sport has its unique nutritional demands.
- Nutrition and physical performance will have three parts, these are: before(pre) exercise, during exercise and after(post) exercise

- What you eat and drink the day before and during the several hours before your workout dictates how much energy you'll have for training and how well you will perform. It also affects how much body fat, glycogen or even muscle tissue you burn.
- Eating the right amount and type of carbohydrate as well as timing your pre-exercise meal correctly will help avoid common problems such as fatigue, dizziness, fainting and stitch.

Why eat before training?

- The main purpose of your pre-workout meal is to stabilise your blood sugar levels during exercise.
- It also staves off hunger and minimises the risk of problems such as stitch and hypoglycaemia (low blood sugar levels).
- Pre-workout meal goal: maximize muscle & liver glycogen stores providing glucose.
- For most regular exercisers, a daily diet providing carbohydrates of around 280–350 g for a 70 kg person

- **Should you train on empty?**
- It is definitely not advisable to train on an empty stomach.
- when your brain isn't getting enough fuel you'll feel faint, lose concentration and risk injury.
- you are more likely to fatigue early as muscle glycogen and blood sugar levels dip.
- you can't expect to exercise very hard or very long when you haven't fuelled your body for several hours.

How much to eat before training

The exact amount you should eat depends on your body weight (heavier people need more) and how hard and long you plan to exercise (eat more for longer, harder workouts).

if you plan to work out for less than 2 hours, aim to eat around 1 g carbohydrate per kg of body weight (or 70 g for a 70 kg person) or 400–600 calories.

For longer workouts or endurance events eat around 2 g carbohydrate per kg of body weight (or 600–800 calories).

- **When to eat before training**
- Ideally, you should aim to have a meal 2–4 hours before a workout. This should leave enough time to partially digest your food although, in practice, the exact timing of your pre-workout meal may depend on your daily schedule.
- Some literature recommended eating a moderately-high Carbohydrate, low fat meal 3 hours before exercise allows you to exercise longer and perform better.

- **What are the best foods to eat just before a workout?**
- Slow-burning or low glycaemic index (GI) foods – that is foods that produce a gradual rise in blood sugar levels are the best foods before a workout.
- low-GI foods help spare muscle glycogen and avoid problems of low blood sugar levels during long training sessions.
- Low-GI meals may also help you burn more fat during exercise

Why drink before training?

- It is important to ensure that you are properly hydrated before training to minimise the risk of dehydration during exercise.
- Exercise feels tougher when you are dehydrated and you cannot train as hard.

When to drink before training?

- The best strategy is to keep hydrated throughout the day rather than load up with fluid just before your workout. Try to make a habit of drinking water regularly. Have a glass of water first thing in the morning and then schedule drinks during your day

Nutrition During Exercise

- Everyone exercising for longer than 30 minutes will certainly benefit from drinking something during exercise. But with the growing array of sports drinks, sports 'waters' and energy drinks it's a confusing choice for most regular exercisers.
- If you plan to exercise longer than 60 minutes, you may also benefit from additional carbohydrate.
- The amount of carbohydrate ingested is important for its contribution to energy expenditure and sparing of liver glycogen

The ideal nutritional strategy during exercise should:

- provide sufficient carbohydrate to maintain blood glucose levels and carbohydrate oxidation
- To provides a fuel source for the brain to maintain skills and decision making, and reduce the perception of fatigue.
- provide water and electrolytes to prevent fluid imbalance
- not cause any gastro-intestinal discomfort
- taste good.

Nutrition after Exercise

- The quicker you can begin refuelling after exercise, the quicker your body will recover.
- Any workout depletes your stores of glycogen – the readily available fuel stored in your muscles – and breaks down muscle tissue.
- Your aim is to rebuild these fuel stores and repair damaged muscle fibers as soon as possible.
- It is during this post-exercise period that your body gets stronger and fitter.

How much to drink?

Start drinking before you even get showered and changed.

The sooner you begin replacing the fluid you have lost through sweat, the sooner you will recover and cut the risk of post-workout dehydration. Fail to drink enough and you will feel listless with a risk of headache and nausea.

you need to drink 750 ml of water for every 0.5 kg of body weight lost during your workout. (1 kilogram of lost weight is equal to 1 litre of sweat, which needs to be replaced with 1.5 litres of fluid).

- **What to drink?**
- If you have exercised for less than an hour, plain water is a good choice followed by a carbohydrate-rich snack within 2 hours.
- For longer or particularly intense workouts, a drink containing carbohydrate (sugar or maltodextrin) and sodium may further speed your recovery.
- carbohydrate at levels of approximately 6 g carbohydrate per 100 ml increases the speed of water absorption into the bloodstream.
- dry fruit juice diluted with an equal volume of water, diluted squash (1 part squash: 6 parts water)

When to eat or drink?

- Whether you are hungry or not, the quicker you consume food or drink after a workout, the quicker your body will recover.
- The enzymes that are responsible for making glycogen are most active immediately after your workout, leaving you a 2-hour window to reload your muscle glycogen.

How to Plan your Training Diet

- Use the Fitness Food Pyramid to devise your daily menu or to check your current eating plan. The foods in the lower layers of the pyramid should form the main part of your diet while those at the top should be eaten in smaller quantities.
- Include foods from each group in the pyramid each day.
- Make sure you include a variety of foods within each group.
- Aim to include the suggested number of portions from each food group each day.

Fitness Food Pyramid

